

Lecture 3 — Load Regulation

Scribe: MIT Student

1 Review

1.1 Method of Assumed States

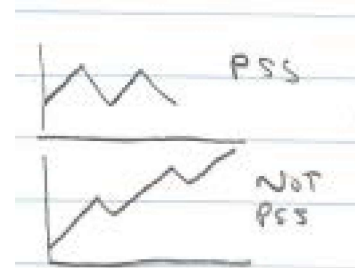
- Assume a state for all switches
- Calculate voltages and currents in the system
- See if any switch conditions are violated
- If not, done. If so, assume a different set of states and try again

1.2 Periodic Steady State

- Converter waveforms repeat every cycle
- In P.S.S.

$$\langle I_C \rangle = \langle C \frac{dV_C}{dt} \rangle = 0$$

$$\langle V_L \rangle = \langle L \frac{dI_C}{dt} \rangle = 0$$



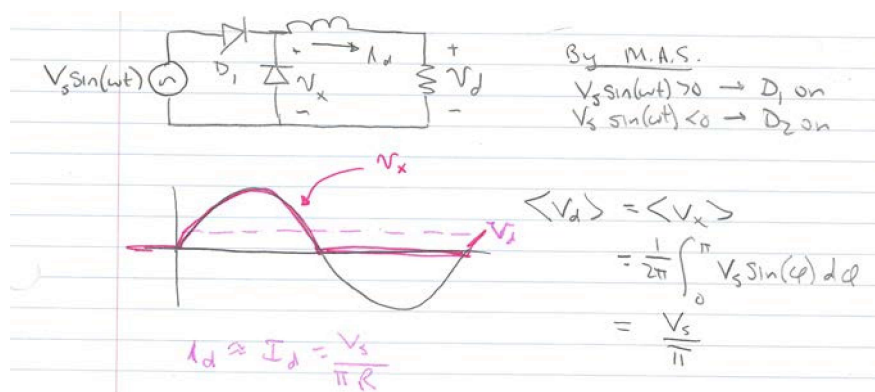
(1)

(2)

2 Rectifiers and DC-Side Characteristics

⇒ Have as show and tell: Alternator, Isolated DC/DC conv.

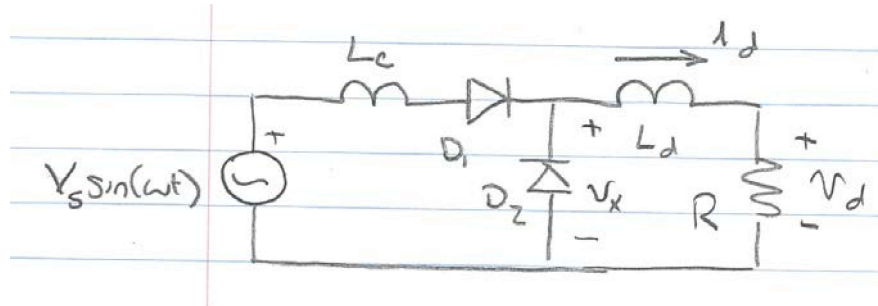
Last time we introduced half a wave rectifier.



Now consider adding some AC-side inductance L_C (reactance $X_C \triangleq \omega L_C$)

This is a relevant issue. L_C can be ...

1. Transformer Leakage Inductance (AC Power or DC-DC)
2. Generator Machine Leakage
3. Line Inductance

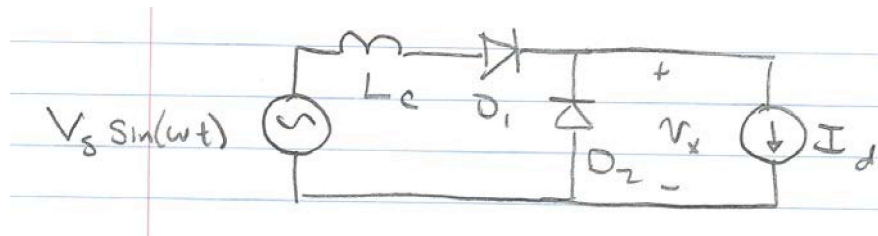


Typically, $L_d \gg L_C$ (filter much bigger than leakage). For simplicity, assume $L_D \rightarrow \infty$ so load looks a current-source.

It is a "special" current source since $\langle V_L \rangle = 0$ in P.S.S.

$$\therefore I_d = \left\langle \frac{V_x}{R} \right\rangle \quad (3)$$

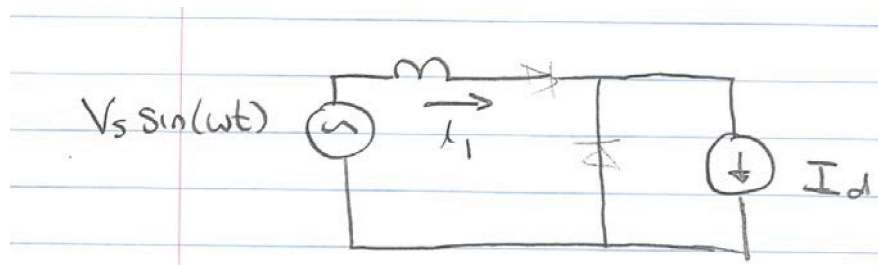
2.1 Simplified Model



Assume we start with D_2 conducting, D_1 off, $V_s \sin \omega t < 0$. What happens when $V_s \sin \omega t$ crosses zero?

- D_1 off is not longer valid ($V_d > 0$) $\therefore D_1$ turns on.
- After turn on I_1 still is zero (no instant I change in L_C) $\therefore D_2$ remains on.

During **commutation**, both devices are on. (NB: commutation means to switch or travel between points.)



$$L_C \frac{dI_1}{dt} = V_s \sin(\omega t) \Rightarrow I_1 = \frac{V_s}{\omega L_C} \int_0^{\omega t} \sin(\omega t) d(\omega t) \quad (4)$$

$$I_1 = \frac{V_s}{\omega L_C} [\cos(0) - \cos(\omega t)] = \frac{V_s}{\omega L_C} [1 - \cos(\omega t)] \quad (5)$$

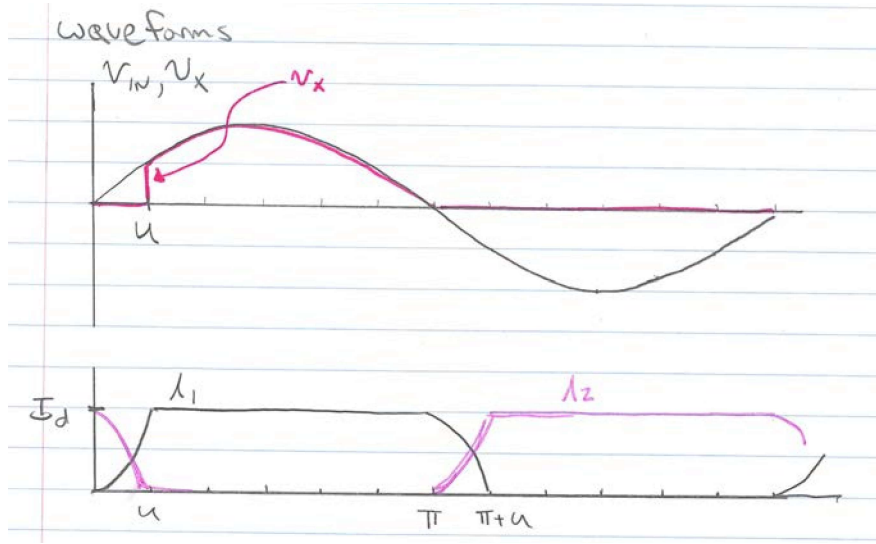
This is valid until some electrical angle $\omega t = u$, when the current through D_2 reaches zero ($I_1 = I_d$)

$$I_d = \frac{V_s}{\omega L_C} [1 - \cos(u)] \quad (6)$$

Commutation interval u such that

$$\cos u = 1 - \frac{\omega L_C I_d}{V_s}$$

The dimensionless term $\frac{\omega L_C I_d}{V_s} = \frac{X_C I_d}{V_s}$ comes up a lot in rectifiers and is called **Reactance Factor**



We lose a piece of V_X (and average output voltage due to commutation (during which V_X is zero). This in turn causes a reduction in filtered output voltage.

$$\begin{aligned} V_O = \langle V_x \rangle &= \frac{1}{2\pi} \int_u^\pi V_s \sin(\varphi) d(\varphi) \\ &= \frac{V_s}{2\pi} [\cos(u) - \cos(\pi)] \end{aligned}$$

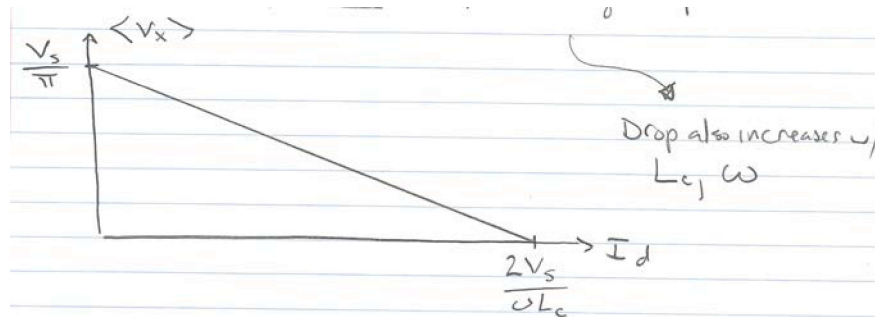
Plugging in for $\cos u$ from our previous calculation.

$$V_O = \langle V_x \rangle = \frac{V_s}{2\pi} \left[2 - \frac{\omega L_C I_d}{V_s} \right]$$

$$V_O = \frac{V_s}{\pi} \left[1 - \frac{1}{2} \frac{\omega L_C I_d}{V_s} \right]$$

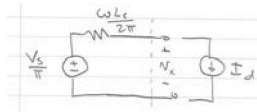
From this we can see:

As load current increases output voltage drops!



At open ckt, we get the same $\langle V_0 \rangle$ as with no L_C . As the current increases, V_0 drops. This is known as **load regulation**, and is typically disliked (we often want constant output voltage independent of load.)

We could model this (in an average sense) as



Note that this is **only** valid in terms of average V-I characteristics. No real dissipation occurs!

Notes: Load regulation is important in many applications.

1. It dominates the performance of automotive alternators
2. It impacts the behavior of DC/DC converters
3. It determines how circuits behave under short- circuit conditions

3 Main Points

1. AC-side reactance in rectifier circuits introduces a **commutation interval** during which multiple devices are on so current can switch between them. This period is u long. For HW rectifier ...

$$\cos u = 1 - \frac{X_C I_d}{V_S}$$

2. The commutation period causes the output voltage to be held low during commutation. The longer the commutation, the lower the output voltage.

⇒ This introduces load regulation of the output.

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6.622 Power Electronics
Spring 2023

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